

Does Lipschitz Boundary plus Small D_I Give Graph Entry?

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1 Question

The proposed input is:

$$\partial E \text{ is Lipschitz,} \quad |\nabla u_E| \text{ is bounded,} \quad D_I(E) := P(E)^2 - 4\pi|E| \ll 1.$$

Can one conclude that E is a small radial graph over a circle?

The answer is:

Not literally from these hypotheses alone. But small D_I gives a strong planar reduction: after discarding satellite components of quadratic area, the convex hull of the main component is a small-amplitude radial graph. To prove that the original set itself is a graph, or that it can be radialized at the sharp torsion cost, one still needs a no-fold/no-fjord lemma.

2 Relevant Known Results

The closest standard results have an extra hypothesis beyond Lipschitz regularity.

1. **Bonnesen in the plane.** For a convex planar set K ,

$$P(K)^2 - 4\pi|K| \geq \pi^2(R_K - r_K)^2.$$

Thus a convex set with small isoperimetric deficit lies in a thin annulus and is a small-amplitude radial graph.

2. **Fuglede.** For convex bodies, or for already nearly spherical sets whose boundary is a small Lipschitz normal graph over the sphere, the isoperimetric deficit controls the graph norm. This is a stability theorem *inside* the convex or graph class; it is not an entry theorem from an arbitrary Lipschitz boundary into the graph class.
3. **Cicalese–Leonardi selection.** The selection principle for the sharp quantitative isoperimetric inequality creates a better competitor with quasiminimality/regularity structure. Again, the regularity comes from the selected variational problem, not from small perimeter deficit alone.

So the available theorem is not “Lipschitz plus small D_I implies graph”. The available theorem is closer to “small D_I plus convexity or near-spherical parametrization gives a quantitative graph estimate”. In our setting, convexification is therefore the natural way to use D_I .

3 Why the literal statement is false

Example 3.1 (Tiny satellite). *Let $0 < r \ll 1$, choose $R_r = (1 - r^2)^{1/2}$, and set*

$$E_r = B_{R_r}(0) \cup B_r(x_0),$$

where the two disks are disjoint and x_0 stays in a fixed bounded region. Then $|E_r| = \pi$ and

$$P(E_r) = 2\pi R_r + 2\pi r = 2\pi + 2\pi r + O(r^2),$$

so

$$D_I(E_r) = P(E_r)^2 - 4\pi^2 = O(r) \rightarrow 0.$$

The boundary is smooth and uniformly Lipschitz in local charts. The torsion gradient is bounded uniformly, since on a disk of radius s the boundary gradient is $s/2$. But E_r is not a radial graph over one circle.

Thus connectedness or a rule for deleting negligible components is necessary. This is not a technicality: small D_I permits small components, and Lipschitz regularity does not remove them.

Remark 3.2. *Even for connected sets, small local Lipschitz constant and small perimeter deficit do not by themselves forbid a shallow overhang inside a thin annulus. Such a set may have several intersections with a ray while paying only a small extra perimeter. Therefore small D_I should be expected to give a graph envelope or a repair theorem, not literal one-crossing for free.*

4 What small D_I does prove in the plane

Normalize $|E| = \pi$ and write

$$P(E) \leq 2\pi + \varepsilon.$$

Here ε is equivalent to $D_I(E)$ in the small regime:

$$D_I(E) = P(E)^2 - 4\pi^2 = (P(E) - 2\pi)(P(E) + 2\pi) \simeq 4\pi\varepsilon.$$

Lemma 4.1 (Satellite components are quadratic). *Let $E \subset \mathbb{R}^2$ have finite perimeter, $|E| = \pi$, and $P(E) \leq 2\pi + \varepsilon$, with ε small. Let E_0 be an indecomposable component of largest area. Then*

$$|E \setminus E_0| \leq C\varepsilon^2.$$

Proof. Let a_j be the areas of the indecomposable components, with a_0 largest, and set $v = \sum_{j \geq 1} a_j$. By the planar isoperimetric inequality and additivity of perimeter over indecomposable components,

$$2\pi + \varepsilon \geq P(E) \geq 2\sqrt{\pi} \left(\sqrt{\pi - v} + \sum_{j \geq 1} \sqrt{a_j} \right).$$

Since $2\sqrt{\pi}\sqrt{\pi - v} \geq 2\pi - Cv$ for small v ,

$$\sum_{j \geq 1} \sqrt{a_j} \leq C\varepsilon + Cv.$$

Using $v \leq (\sum_{j \geq 1} \sqrt{a_j})^2$ and absorbing for small ε gives $v \leq C\varepsilon^2$. □

Lemma 4.2 (Convex graph envelope). *Let $F \subset \mathbb{R}^2$ be the largest component from Lemma 4.1, and assume F is a continuum. Let*

$$K = \text{co}(F).$$

Then $P(K) \leq P(F) \leq 2\pi + C\varepsilon$, and after translating and rescaling K to area π , one has

$$K^\# = \{z + re^{i\theta} : 0 < r < 1 + \varphi_K(\theta)\}, \quad \|\varphi_K\|_{L^\infty} \leq C\sqrt{\varepsilon}.$$

Proof. For planar continua, perimeter does not increase under convexification:

$$P(\text{co}(F)) \leq P(F).$$

The area of K is bounded and close to π up to a harmless $O(\varepsilon)$ error in the small-deficit regime. Bonnesen's inequality for convex planar sets gives

$$P(K)^2 - 4\pi|K| \geq \pi^2 (R_K - r_K)^2,$$

where R_K and r_K are the circumradius and inradius of K with respect to a suitable center. Since the left side is $O(\varepsilon)$,

$$R_K - r_K \leq C\sqrt{\varepsilon}.$$

After area normalization, this is exactly containment in an annulus of thickness $C\sqrt{\varepsilon}$. A convex body containing the center as an interior point is a radial graph, and the annular bound gives the stated L^∞ amplitude estimate. $\square$

5 Where Lipschitz and $|\nabla u|$ help

The previous two lemmas did not use Lipschitz boundary or the gradient bound. Those hypotheses are still useful, but they play a different role.

1. A uniform Lipschitz boundary gives local finite-overlap charts and prevents arbitrarily wild fractal boundary behavior.
2. A uniform bound for $|\nabla u_E|$ prevents high-gradient boundary defects, such as tiny holes for the torsion problem.
3. In the selected-level setting, the stronger estimate

$$\| |\nabla u_E| - c \|_{L^\infty(\partial E)} \ll 1$$

is much more powerful than mere boundedness. It rules out small components and low-gradient tentacles that small D_I alone permits.

But these hypotheses still do not give the missing radial projection injectivity. Local Lipschitz graphs can fold relative to the radial projection.

6 Correct theorem to prove

The useful replacement for literal graph entry is the following.

Proposition 6.1 (What the current inputs reduce to). *Assume the stopped planar level E satisfies*

$$|E| = \pi, \quad P(E) \leq 2\pi + \varepsilon, \quad \partial E \text{ uniformly Lipschitz},$$

and has the selected-level torsion bounds

$$\|\nabla u_E\|_{L^\infty} \leq L, \quad \| |\nabla u_E| - c \|_{L^\infty(\partial E)} \leq \eta.$$

Then small D_I gives, quantitatively, a small-amplitude convex radial graph envelope $K^\#$ for the main component of E , after deleting satellites of area $O(\varepsilon^2)$.

What remains to close the hybrid route is the sharp radialization estimate

$$|E\Delta G| \leq C\sqrt{\delta}, \quad \delta_T(G) \leq C\delta,$$

where G is a small-amplitude radial graph. The new work is to prove that the parts of $K \setminus E$ and $E \setminus K$ coming from folds, fjords, and outward fingers can be filled or pruned at this torsion cost. This is where the D_H thin-finger estimate and the capacity/no-fjord mechanism must enter.

7 Verdict

There is a standard route from small D_I to a graph envelope: component cleanup, convex hull, and Bonnesen. It is fully quantitative in dimension two and gives an L^∞ radial graph after convexification.

There is not, however, a theorem saying that a uniformly Lipschitz set with bounded torsion gradient and small D_I is itself a radial graph. The literal statement is false without at least deleting negligible components, and for connected sets it still needs a no-fold/no-fjord argument. Thus the right next lemma is not “ D_I small implies graph”; it is:

$$\text{convex graph envelope} + \text{Serrin}/D_H \text{ control} \implies \text{sharp radialization}.$$